

Computer Graphics Programming Project

Title: Implementation of Fundamental Computer Graphics Algorithms

Objective

Develop a complete computer graphics application that demonstrates the implementation of the essential algorithms used in 2D computer graphics. The project should allow users to interactively draw, transform, and manipulate graphical objects.

Requirements

Part 1: Drawing Algorithms

Implement the following line and circle drawing algorithms:

- Digital Differential Analyzer (DDA) Line Drawing Algorithm
- Bresenham's Line Algorithm
- Midpoint Circle Algorithm
- Bresenham's Circle Algorithm

Functionalities

- Draw lines using both DDA and Bresenham algorithms.
 - Draw circles using Midpoint and Bresenham algorithms.
 - Compare the output and efficiency of each algorithm.
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Part 2: 2D Transformations

Implement the following geometric transformations:

- Translation
- Rotation
- Scaling
- Reflection
- Shearing

Functionalities

- Apply transformations to basic shapes such as lines, triangles, rectangles, and polygons.
 - Allow multiple transformations in sequence.
 - Display original and transformed objects simultaneously.
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Part 3: Clipping Algorithms

Implement the following clipping techniques:

- Cohen–Sutherland algorithm for line clipping
- Sutherland–Hodgman algorithm

Functionalities

- Clip lines and polygons against a rectangular clipping window.
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Part 4: Curves

Implement:

- Bezier curve
 - B-spline
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Technical Specifications

- Programming Language: C++, Python, Java, or C#
 - Graphics Library: OpenGL
 - User Interface: Menu-driven or GUI-based
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Deliverables

1. Complete source code

2. Executable program
 3. Project report including:
 - Algorithm description
 - Mathematical equations
 - Screenshots
 - Performance analysis
 4. Presentation slides
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Bonus Features

- Interactive mouse input
 - Animation of transformations
 - Performance comparison charts
 - Save and load drawings
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Submission Guidelines

- Submit the source code in a compressed ZIP file.
 - Include a README file with compilation and execution instructions.
 - Submit the report in PDF format.
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Expected Output

The application should provide:

- Interactive drawing area
- Algorithm selection menu
- Transformation controls
- Clipping visualization

- Performance comparison results
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Learning Outcomes

After completing this assignment, students will be able to:

- Understand raster graphics algorithms.
- Apply geometric transformations.
- Perform clipping operations.
- Generate curves and filled regions.
- Develop interactive graphics applications.